



# Ankylosing Spondylitis

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Ankylosing spondylitis is a [spondyloarthritis](#) characterized by inflammation of the spine (spondylitis), large joints, and fingers and toes, resulting in stiffness and pain.

- Prolonged joint pain, morning back stiffness, and eye inflammation are common.
- The diagnosis is based on symptoms, x-rays, and established criteria.
- Nonsteroidal anti-inflammatory drugs and sometimes [sulfasalazine](#) or [methotrexate](#) can help relieve the arthritis in limbs.
- Medications that inhibit tumor necrosis factor and some other biologic agents are very effective for spine and limb arthritis.

Ankylosing spondylitis is 3 times more common among men than women, developing most commonly between the ages of 20 and 40. The cause of ankylosing spondylitis is not known, but the disease tends to run in families, indicating that genetics plays a role. Ankylosing spondylitis is 10 to 20 times more common among people whose parents or siblings have it. The *HLA-B27* gene is present in 90% of White people who have ankylosing spondylitis, but it is also present in up to 10% of the general population depending on ethnicity. However, having the *HLA-B27* gene does not mean that a person has or will develop ankylosing spondylitis. For example, only 50% of identical twins both have the disorder, which suggests that unknown environmental factors also may be involved.

## Ankylosing Spondylitis

3D MODEL



## Symptoms of Ankylosing Spondylitis

In ankylosing spondylitis, mild to moderate flare-ups of inflammation may alternate with periods of almost no symptoms.

The most common symptom is

- Back pain

Back pain varies in intensity from one episode to another and from one person to another. Pain is often worse during the night and in the morning. Early morning stiffness that is relieved by activity is also very common. Pain in the lower back and the associated muscle spasms are often relieved by bending forward. Therefore, untreated people often develop a stooped posture, which can get worse and become permanent. In others, the spine becomes noticeably straight and stiff.

Occasionally, loss of appetite, weight loss, excessive fatigue, and anemia can accompany the back pain. If the joints connecting the ribs to the spine are inflamed, the pain may limit the ability to expand the chest to take a deep breath. Stiffness (fusion) of the spine can restrict the ability to expand the chest wall as well.

Occasionally, pain starts in large joints, such as the hips, knees, and shoulders.

Approximately 25% of people have bodywide (systemic) symptoms:

- Recurring attacks of painful but sometimes mild eye inflammation ([uveitis](#)) is common, but usually does not impair vision if treated promptly.
- In a few people, inflammation of a heart valve results in a permanently damaged valve, or other problems can affect the heart or aorta.
- If damaged vertebrae press against nerves or the spinal cord, numbness, weakness, or pain can develop in the area supplied by the affected nerves. [Cauda equina syndrome](#) is an occasional complication when nerves coming out of the lower part of the spinal cord are affected.
- Lung disorders such as cough, coughing up blood (hemoptysis), and shortness of breath (dyspnea) can rarely develop.
- [Achilles tendinitis](#), [plantar fasciitis](#), and patellar tendinitis can develop. Dactylitis (sausage digits), which is painful swelling of a finger or toe, can also develop.

## Dactylitis

IMAGE



This photo shows one finger that has dactylitis (left) and the same finger on the opposite hand that does not (right).

IMAGE COURTESY OF KINANAH YASEEN, MD.

## Diagnosis of Ankylosing Spondylitis

- X-rays
- Blood tests
- Sometimes magnetic resonance imaging (MRI)
- Established criteria

The diagnosis of ankylosing spondylitis is based on the pattern of symptoms, a family history of the disorder, and on x-rays of the spine, pelvis, and affected joints. [X-rays](#) usually, but not always, show a wearing away (erosion) of the joint between the spine and the hip bone (sacroiliac joint) and the formation of bony bridges between the vertebrae, making the spine stiff (sacroiliitis). In some people, sacroiliitis is not visible on x-rays but may be detected by [MRI](#) of the pelvis or spine.

Blood tests are done to determine the erythrocyte sedimentation rate (ESR), a test that measures the rate at which red blood cells settle to the bottom of a test tube containing blood, and also to determine the level of C-reactive protein and sometimes the presence of the *HLA-B27* gene. High levels of ESR and C-reactive protein indicate inflammation but may not indicate the severity of the disorder. People may have the *HLA-B27* gene and not have spondylitis. The presence of the *HLA-B27* gene increases the likelihood of ankylosing spondylitis but does not confirm the diagnosis.

### LAB TEST

[Erythrocyte Sedimentation Rate \(ESR\)](#)



Doctors can also base the diagnosis on an established set of criteria, but there are several different sets of criteria and some of them are undergoing changes. For example, the following criteria are sometimes applied to people who have had back pain for more than 3 months and who are under 45 years of age when their symptoms start.

There are 2 parts to the criteria described here: imaging (x-rays or MRI) criteria and clinical (examination and blood test) criteria. People who fulfill one or both parts of the criteria may have ankylosing spondylitis.

To fulfill the imaging part of the criteria, people must have sacroiliitis confirmed by x-rays or MRI and at least 1 of the features in the list below. To fulfill the clinical part of the criteria, people must have the *HLA-B27* gene and at least 2 of the features in the list below:

- Dactylitis (swelling of an entire finger or toe)
- Painful inflammation of the heel
- Family history of spondyloarthritis
- History of inflammatory back pain
- Arthritis
- [Psoriasis](#)
- [Inflammatory bowel disease](#)
- Eye inflammation ([uveitis](#))
- A high level of C-reactive protein
- Relief of pain and inflammation with nonsteroidal anti-inflammatory drugs ([NSAIDs](#))

Back pain due to inflammation usually begins gradually when a person is 40 or younger. People have stiffness in the morning that is relieved by movement.

## Treatment of Ankylosing Spondylitis

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- Nonsteroidal anti-inflammatory drugs
- Tumor necrosis factor (TNF) inhibitors, interleukin-17 (IL-17) inhibitors, or Janus kinase (JAK) inhibitors
- Exercise

Treatment of ankylosing spondylitis is focused on

- Relieving back and joint pain
- Maintaining range of motion in the joints
- Preventing disease complications and structural damage in other organs
- Preventing or correcting spinal deformities

[Nonsteroidal anti-inflammatory drugs](#) (NSAIDs) can reduce pain and inflammation, thus enabling people to do important exercises to retain posture, including stretching and deep breathing. Biologic medications are typically used when NSAIDs do not adequately control ankylosing spondylitis.

TNF inhibitors (for example, [etanercept](#), [adalimumab](#), [infliximab](#), [golimumab](#), and [certolizumab pegol](#)) often effectively relieve back pain and inflammation. [Adalimumab](#) or [infliximab](#) is preferred over [etanercept](#) for the treatment of ankylosing spondylitis in people with recurring [uveitis](#).

IL-17 inhibitors (for example, [secukinumab](#), [ixekizumab](#)) are other biologic agents that can also reduce inflammation and joint symptoms.

JAK inhibitors (for example, [upadacitinib](#) and [tofacitinib](#)) are targeted synthetic disease-modifying antirheumatic drugs (DMARDs) that may be used in people whose symptoms were not relieved with NSAIDs, a TNF inhibitor, or an IL-17 inhibitor.

The long-range goals of ankylosing spondylitis treatment are to maintain proper posture and develop strong back muscles. Daily exercises strengthen the muscles that oppose the tendency to bend and stoop. It has been suggested that people spend some time each day—often while reading—lying on their stomach propped up on their elbows because this position extends the back and helps to keep the back flexible. In addition to exercise, people who smoke cigarettes should be encouraged to stop because it can make the arthritis worse.

Steroid eye drops and dilating eye drops may help in the short-term treatment of inflammation of the eyes that comes and goes, and an occasional steroid injection may be helpful for 1 or 2 joints other than the spine. Muscle relaxants and opioid analgesics are occasionally used, but for only brief periods to relieve severe pain and muscle spasms.

If the hips become eroded or fixed in a bent position, surgical treatment to replace the joint can relieve pain and restore function.

## Prognosis for Ankylosing Spondylitis

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Most people with ankylosing spondylitis develop some disabilities but can still lead normal, productive lives. In some people, the disease is more progressive, causing severe deformities. The prognosis for ankylosing spondylitis is discouraging for people who develop extreme stiffness of the spine or hip, [uveitis](#), or heart or lung disease.

## More Information

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The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

[Arthritis Foundation](#)

### Drug Information for the Topic

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